

Methods

Preview

1. Data: acoustic/articulatory data from published datasets
2. Distinctiveness measurement in acoustics & articulation
3. Bigram surprisal calculation
4. Phoneme flexibility: degree of freedom between articulatory synergies & the speech target



Methods

Speech flexibility

- Flexibility of each phoneme type was estimated as its conditional entropy through mixture density networks (MDN; cf. Richmond 2006; Zen & Senior 2014).
- MDN predicts a K-mixture (K=5) probability density distribution based on the input conditions.
- In this study:
 - $P(\text{observed articulatory matrix} \mid \text{acoustic matrix})$
 - MDNs were first trained on the train sets & used to estimate entropies for the test sets.